

13. Top-Down Parsing to Check Grammar

CODE

```
#include <stdio.h>
```

```
#include <string.h>
```

```
char str[100];
```

```
int pos = 0;
```

```
int S()
```

```
{
```

```
    if (str[pos] == 'a')
```

```
    {
```

```
        pos++;
```

```
        if (!S())
```

```
            return 0;
```

```
        if (str[pos] == 'b')
```

```
        {
```

```
            pos++;
```

```
            return 1;
```

```
        }
```

```
        return 0;
```

```
    }
```

```
    return 1;
```

```
}
```

```
int main()
{
    printf("Grammar: S -> aSb | epsilon\n");
    printf("Enter string: ");
    scanf("%s", str);

    if (S() && str[pos] == '\0')
        printf("String is ACCEPTED.\n");
    else
        printf("String is REJECTED.\n");

    return 0;
}
```

OUTPUT

```
Grammar: S -> aSb | epsilon
Enter string: aabb
String is ACCEPTED.

-----
Process exited after 17.71 seconds with return value 0
Press any key to continue . . . |
```